CLINICAL TEACHING NOTE | FAMILY MEDICINE

Migraine

Assessment and management in primary care

1. Diagnostic approach

Step 1: Classify by ICHD-3 criteria

- Migraine without aura: recurrent attacks lasting 4–72 hours, typically unilateral and pulsating, moderate–severe, aggravated by routine activity, with nausea and/or photophobia and phonophobia.
- Migraine with aura: fully reversible visual, sensory, or speech symptoms that precede or accompany the headache, developing over at least 5 minutes and lasting 5–60 minutes.
- Chronic migraine: headache on ≥15 days a month for more than 3 months, with migrainous features on at least 8 of those days.
- Medication overuse headache: headache on ≥15 days a month in a patient overusing acute treatment (triptans or combination analgesics on ≥10 days/month, or simple analgesics on ≥15 days/month) for more than 3 months.

Step 2: Key history

- Headache characteristics: onset, laterality, quality, severity, duration, frequency, and triggers such as stress, sleep disturbance, hormonal change, diet, or screen time.
- Associated symptoms: nausea or vomiting, photophobia, phonophobia, and any aura symptoms (visual scintillations or scotoma, paraesthesia, dysphasia).
- Premonitory and postdromal symptoms: mood change, yawning, food cravings, fatigue.
- Menstrual pattern in women, and use of hormonal contraception or HRT.
- Frequency and type of acute medication use, to identify a risk of medication overuse headache.
- Family history of migraine, and SNOOP red flags: Systemic symptoms, Neurological signs, sudden Onset, Older age at onset (>50), Pattern change or Positional, Precipitated by valsalva, Papilloedema, Progressive or atypical course, pregnancy/postpartum, or Painful eye with autonomic features.

Step 3: Examination

- Full neurological exam including cranial nerves, fundoscopy for papilloedema, and gait/cerebellar assessment.
- Blood pressure, to exclude a hypertensive contribution.
- Temporal artery palpation and inspection in patients over 50, to exclude giant cell arteritis.
- Neck examination for a cervicogenic contribution, and assessment of the temporomandibular joints and sinuses.
- Look specifically for features of secondary headache: focal deficit, altered consciousness, or meningism.

2. Distinguishing the common causes

	Migraine	Tension-type headache	Cluster headache	Secondary headache (red flag)
Timing / duration	Attacks of 4–72 hours	30 minutes to 7 days	15–180 minutes, several attacks/day, clustering over weeks	Sudden/thunderclap, or progressive over days–weeks

	Migraine	Tension-type headache	Cluster headache	Secondary headache (red flag)
Quality / location	Unilateral, throbbing, moderate–severe	Bilateral, band-like pressure, mild–moderate	Severe, unilateral periorbital or temporal	"Worst headache of life" or new, unfamiliar pattern
Associated features	Nausea, photophobia/phonophobia, ± aura	No nausea; at most mild photo- or phonophobia	Ipsilateral lacrimation, ptosis, miosis, nasal congestion; restlessness	Focal neurological deficit, fever, papilloedema, altered consciousness
Triggers	Stress, hormonal change, sleep disturbance, diet	Stress, poor posture, fatigue	Alcohol during a cluster bout; circadian pattern	Valsalva, exertion, positional change
Course	Episodic; may evolve into chronic migraine	Episodic or chronic, not disabling	Episodic clusters with remission, or chronic	Progressive — needs urgent work-up

3. Investigations

Migraine is a clinical diagnosis; investigations are reserved for excluding secondary causes, not for routine confirmation:

- No imaging is needed for typical migraine with a stable pattern and a normal examination.
- Neuroimaging (MRI preferred) if red flags are present: new or changed headache pattern, abnormal neurological exam, onset after age 50, thunderclap onset, progressive course, or atypical aura.
- ESR/CRP if giant cell arteritis is suspected (age over 50, jaw claudication, visual symptoms, temporal tenderness).
- Consider lumbar puncture if subarachnoid haemorrhage is suspected with a normal CT, or if idiopathic intracranial hypertension is suspected (with opening pressure measurement).
- A headache diary to characterise pattern, triggers, and acute medication use before starting preventive therapy.

Red flags: same-day or emergency referral

- Thunderclap headache reaching maximal intensity within seconds to minutes — exclude subarachnoid haemorrhage.
- New headache with fever, neck stiffness, or altered consciousness — exclude meningitis or encephalitis.
- New focal neurological deficit, seizure, or papilloedema.
- New headache in pregnancy or postpartum with hypertension — exclude pre-eclampsia or eclampsia.
- New headache onset after age 50, or in a patient with cancer or immunosuppression.
- Headache triggered by valsalva, exertion, or positional change, or with visual loss and jaw claudication in a patient over 50 (giant cell arteritis).

4. Management

General measures for all causes

- Use a headache diary to identify triggers and monitor both treatment response and acute medication use.
- Lifestyle measures: regular sleep and meals, adequate hydration, limiting caffeine, stress management, and regular exercise.
- Limit acute medication use to fewer than 2–3 days a week to avoid medication overuse headache.

- Explain the diagnosis and reassure regarding the benign nature of migraine once red flags have been excluded.

Acute treatment

Line	Drug	Typical dose	Key cautions
First line	Oral triptan (e.g. sumatriptan)	50–100 mg at onset; may repeat once after 2 hours (max per product)	Avoid in ischaemic heart disease, uncontrolled hypertension, or stroke history; do not combine with ergotamine.
First line	NSAID (e.g. ibuprofen, naproxen)	Standard analgesic dose at symptom onset	GI, renal, and cardiovascular caution; avoid frequent use.
First line	Antiemetic (e.g. metoclopramide, prochlorperazine)	Standard dose, oral or IM	Extrapyramidal effects, particularly in young women; used adjunctively for nausea and to aid drug absorption.
Second line	Combination triptan + NSAID	As per individual components	Same cautions as the individual components.

Preventive treatment

Consider if attacks occur on ≥4 days a month, are disabling, or there is a risk of acute medication overuse.

Line	Drug	Typical dose	Key cautions
First line	Propranolol	80–240 mg/day	Avoid in asthma, heart block, or bradycardia.
First line	Amitriptyline	10–75 mg at night	Anticholinergic effects; use cautiously in the elderly and in cardiac disease.
Second line	Topiramate	Slow titration to 50–100 mg/day	Cognitive slowing, paraesthesia, weight loss; teratogenic — avoid without reliable contraception.
Second line	Candesartan	8–32 mg/day	Monitor renal function and potassium.
Specialist	CGRP monoclonal antibody or gepant	Per specialist protocol	Specialist initiation; cost and access considerations.

Medication overuse headache

- Advise withdrawal of the overused acute medication (abrupt or tapered depending on the drug); bridging therapy may be needed.
- Address the underlying migraine with preventive therapy, and refer if withdrawal proves difficult or the diagnosis is uncertain.

Menstrual migraine

- Consider short-term mini-prophylaxis (e.g. NSAID or triptan) around the perimenstrual window.
- Consider hormonal strategies in conjunction with gynaecology where appropriate.

5. Follow-up

- Review the headache diary and treatment response 4–8 weeks after starting acute or preventive treatment.
- Reassess the frequency of acute medication use at every review to prevent medication overuse headache.
- Adjust preventive therapy if response remains inadequate after an adequate trial, usually 8–12 weeks at target dose.

- Revisit the diagnosis if new features develop, especially a change in pattern or the emergence of red flags.

Key take-home points

- Diagnose migraine using ICHD-3 criteria; imaging is not routinely needed in the absence of red flags.
- Screen for SNOOP red flags at every new or changed presentation.
- Limit acute medication to fewer than 2–3 days a week to prevent medication overuse headache.
- Offer preventive treatment for frequent or disabling attacks; propranolol and amitriptyline are first line.
- A headache diary supports diagnosis, monitoring, and treatment decisions.

References

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This summary is an educational aid compiled from standard family medicine and specialty references; it is not a substitute for local clinical guidelines or the primary literature.